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Palaeomagnetism and magnetostratigraphy of uppermost Permian strata, southeast New Mexico, USA: correlation of the Permian–Triassic boundary in non-marine environments

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SUMMARY

Continental red sandstone and siltstone rocks of the Dewey Lake (Quartermaster) Formation at Maroon Cliffs, near Carlsbad, New Mexico, are characterized by two components of magnetization with partially overlapping laboratory unblocking temperature spectra. Both magnetizations display high coercivities (>100 mT), probably residing in haematite. A north-directed magnetization with steep positive inclination unblocks between 100 and 650 °C, isolating a predominantly northwest-directed magnetization, with shallow inclination, of near uniform normal polarity and maximum unblocking temperatures of 680 °C.

We collected samples from 24 palaeomagnetic sites (i.e. individual beds) from a $\sim$60 m thick section of flat-lying strata disconformably overlying carbonate and evaporite rocks of the Rustler Formation. The upper member of the Rustler Formation contains a Late Permian (early Changxingian) marine invertebrate and conodont fauna. Of the sampled sites, four yield only steep magnetizations, interpreted to be recent overprints. Eight sites did not yield well-grouped site means and were excluded from the final calculations. The formation mean (dec $= 337.7°$, inc $= 9.2°$; $k = 31.6$, $\alpha_{95} = 7.8°$, $N = 12$ sites) defines a palaeomagnetic pole located at 55.2°N, 117.5°E, in good agreement with other Late Permian North American cratonic poles.

Correlation of the short polarity sequence of this section of Dewey Lake strata is unambiguous. Compared with the polarity stratigraphy of marine sections in Asia, and supported by isotopic age determinations on a widespread bentonite bed in Dewey Lake strata in west Texas (approximately 251 Ma) and fossil data for the underlying Rustler Formation, the magnetostratigraphy is consistent with deposition of the Dewey Lake Formation during the latest Changxingian (Late Permian) stage.

Key words: magnetostratigraphy, palaeomagnetism, Permian.

INTRODUCTION

In southeastern New Mexico and west Texas, Upper Permian strata have been used to define the Guadalupian and Ochoan stratotypes of the Permian regional chronostratigraphic scale in North America. Intra-Ochoan unconformities suggest that Ochoan strata represent an incomplete record of Late Permian times (Lucas & Anderson 1994, 1997). The term Ochoa Group is thus used in southwestern North America for a succession of marine and non-marine strata that roughly corresponds to the non-marine Tatarian succession of the Upper Permian of the Russian platform and the lowermost Zechstein–Buntsandstein succession of northwest Europe, which is also dominated by evaporites and non-marine strata.

Ochoan red beds of the Dewey Lake Formation in the southern High Plains of the North American craton are characterized by normal- and reverse-polarity magnetizations, indicating that they postdate the Kiaman Superchron (Molina-Garza *et al.* 1989). Correlation with marine biochronology suggests that the Ochoan spans the Longtanian and Changxingian stages of the Late Permian (Lucas & Anderson 1994, 1997). Recent single-crystal ^{40}Ar–^{39}Ar and SHRIMP zircon analyses of a single ash layer interbedded in strata of the Dewey Lake Formation indicate an isotopic age of about

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251 Ma (Renne *et al.* 1996). This age estimate is identical to that determined from high-precision geochronology in the proposed marine Permian–Triassic boundary stratotype at Meishan, Sichuan, south China (Renne *et al.* 1996; Bowring *et al.* 1998; Ross *et al.* 1994), suggesting that Dewey Lake strata may contain a record of the Palaeozoic–Mesozoic erathem boundary. Here, we report palaeomagnetic and magnetostratigraphic data for a 60 m thick section of the Dewey Lake Formation (uppermost Permian, Ochoan) in southeastern New Mexico, USA.

GEOLOGICAL SETTING

The Dewey Lake Formation consists of a thin succession of non-marine, weakly haematite-cemented, light red sandstone, siltstone, and mudstone, deposited north and northwest of the Permian Basin (Schiel 1988). These rocks are exposed intermittently in southeastern New Mexico and west Texas but have an extensive subsurface distribution (Fig. 1, inset). The name Quartermaster is extensively used for the outcrops of stratigraphically equivalent Ochoan red beds in the Texas Panhandle and Oklahoma. In southeastern New Mexico, the Dewey Lake Formation is between 30 and 115 m thick. It is the youngest formation of the Ochoa Group (Lucas & Anderson 1994). The Dewey Lake overlies evaporites and carbonate rocks of the Rustler Formation and is disconformably overlain by Upper Triassic strata of the Santa Rosa Formation (Lucas & Anderson 1994). Three molluscan taxa from the Dewey Lake Formation in Texas suggest that it is of Permian age. Other biostratigraphic data are lacking, but the underlying rocks are of Late Permian age (Lucas & Anderson 1994). The upper contact of the Dewey Lake Formation is an erosional unconformity with variable amounts of relief. The Santa Rosa Formation can be physically correlated with the lower part of the Dockum Group in west Texas, where its Late Triassic age is well established (Hunt & Lucas 1991; Lucas & Anderson 1993).

SAMPLING AND METHODS

For this study, we collected six to 10 orientated samples from each of 24 palaeomagnetic sites (one site = one stratigraphic layer) spaced at irregular intervals in a flat-lying section of the Dewey Lake Formation that crops out in the Maroon Cliffs, about 20 km east of Carlsbad, New Mexico (Fig. 1). Here, the rocks are well exposed and form prominent cliffs along the Pecos River escarpment. Samples were collected with a portable drill and orientated with magnetic and sun compasses. At this locality, the Dewey Lake Formation consists of mudstone and siltstone intercalated with fine- to coarse-grained sandstone layers 0.2–0.7 m thick, with the abundance of sandstone reflecting relative proximity to the basin edge. Samples were collected primarily from well-indurated sandstone layers.

In the laboratory, one or two specimens from each sample were prepared for NRM (natural remanent magnetization) measurements in a 2-G Enterprises 760-R cryogenic magnetometer. All specimens were subjected to stepwise AF (alternating field), thermal, or chemical demagnetization. AF demagnetization used a coil system in-line with the cryogenic magnetometer capable of reaching peak inductions of 130 mT. Thermal demagnetization up to 690 °C was performed in 10 to 20 steps using a Schonstedt TSD-1 furnace. For chemical demagnetization, samples were prepared using the methods described by Henry (1979). Samples were subject to leaching using concentrated hydrochloric acid (37 per cent) at room temperature for up to 30 hr in a magnetic shield with a residual field of less than 50 nT. Vector components to the remanent magnetization were identified from visual inspection of demagnetization diagrams (Zijderveld 1967) and directions were calculated using principal component analysis (PCA) (Kirschvink 1980). Site means and the overall formation mean were calculated using Fisher statistics (Fisher 1953).

DEMAGNETIZATION RESULTS

The remanent magnetization of medium- to fine-grained sandstones of the Dewey Lake Formation is of moderate intensity

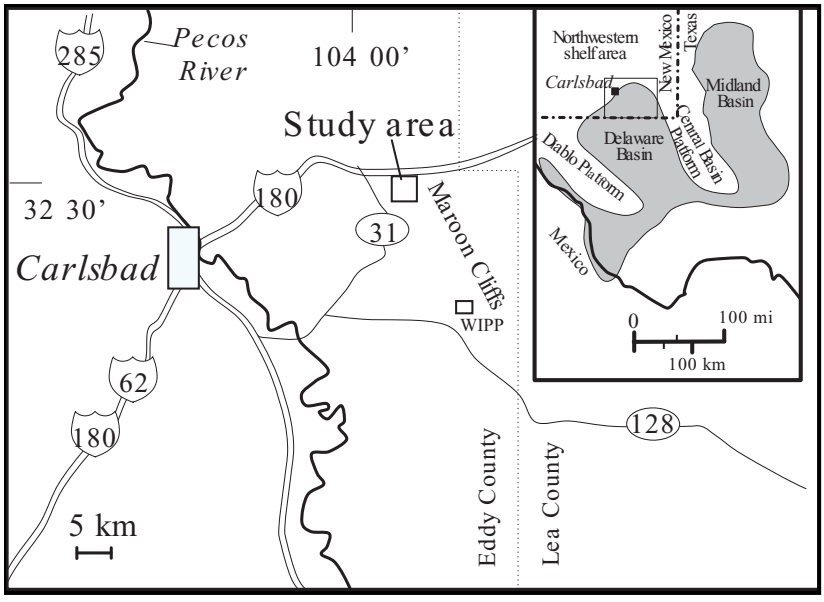


Figure 1. Map of the Carlsbad region of southeast New Mexico showing the location of the study area. The inset shows the location of the Carlsbad region in a map of the Permian basin of west Texas.

(10^{-3}–10^{-4} A m^{-1}) and it normally consists of two vector components (Fig. 2a). A north-directed and steep positive magnetization of distributed laboratory unblocking temperatures overprints a typically northwest-directed and shallow characteristic magnetization (ChRM). The ChRM unblocks over a narrow range of temperatures between about 630 and 690 °C. Normally, both components exhibit high coercivity (> 100 mT; Fig. 2b) and probably reside in haematite, but there are sites where the overprint may partially reside in magnetite (Fig. 3). The steep north-directed magnetization is interpreted as a secondary magnetization of relatively recent origin. Chemical demagnetization (Fig. 2c) preferentially removes the steep, north-directed overprint, and the ChRM is isolated only after haematite pigment has been completely dissolved, suggesting that fine-grained haematite is the principal carrier of the secondary remanence. Unfortunately, calcite is a more dominant cementing agent, precluding the use of chemical demagnetization for long periods of time.

In samples with well-developed overprints (for example sites mcf15 and mcf19) and sites that did not yield characteristic components (for example mcf3 and mcf12) that were subjected to chemical demagnetization for 30 hr, a small, north-directed magnetization was removed with subsequent AF treatment. The rock-magnetic experiments shown below demonstrate that in these sites the north-directed overprint resides in both fine-grained haematite and magnetite.

In about 30 per cent of the demagnetized samples, the secondary magnetization is easily removed, and PCA calculations of directions for the ChRM yield excellent linear fits, with MAD (maximum angular deviation) values below 5° for unanchored line segments. In other cases, the ChRM is difficult to isolate because complete removal of the overprinting magnetization requires heating to a temperature of about 650 °C, and above that temperature demagnetization behaviour typically becomes erratic, possibly due to alteration during heating but also due to decreased magnetic moments (Fig. 2d). Overall, the average MAD value of samples included in site mean calculations is 9.4°; samples with MAD of more than 15° were excluded from the calculations. The characteristic magnetization was isolated in 78 samples from 20 sites. In the remaining samples the north-directed overprint dominates the NRM.

Most ChRM directions are northwest-directed with shallow inclination (normal polarity; Fig. 4). Samples with reverse-polarity magnetizations (principally from site mcf13 but also from site mcf14) are interpreted as records of a short duration interval of reverse polarity (Figs 2e and f). Although the evidence for reverse-polarity magnetizations is unequivocal, the record is complex because site mcf14 samples yield mixed polarity results, and several samples from site mcf13 yield intermediate directions (Fig. 2f). These observations can be explained in two ways. The unblocking temperature spectra

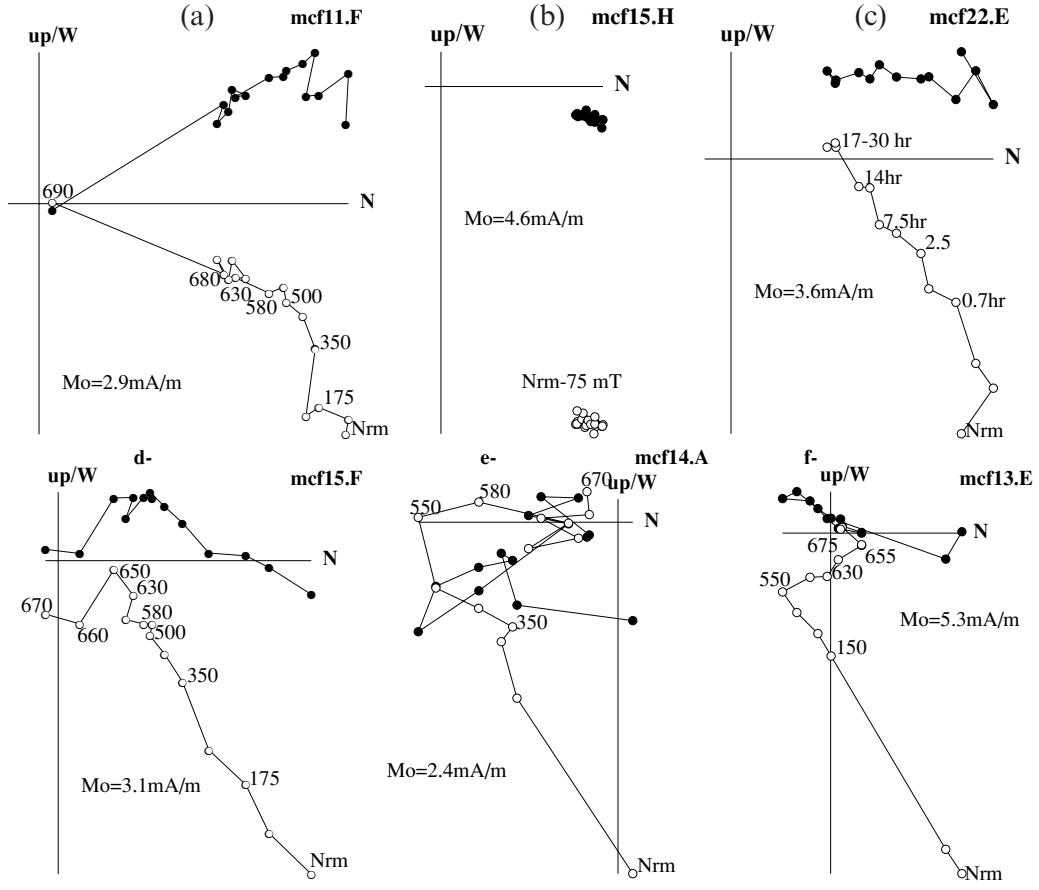


Figure 2. Orthogonal demagnetization plots (Zijderveld 1967) of selected samples of the Dewey Lake Formation. The most effective method for isolating a characteristic magnetization is thermal demagnetization. Alternating field and chemical demagnetization are shown in (b) and (c), respectively. Solid (open) symbols are projections on the horizontal (vertical) plane. The samples illustrated in (e) and (f) are interpreted as having magnetizations of reverse polarity.

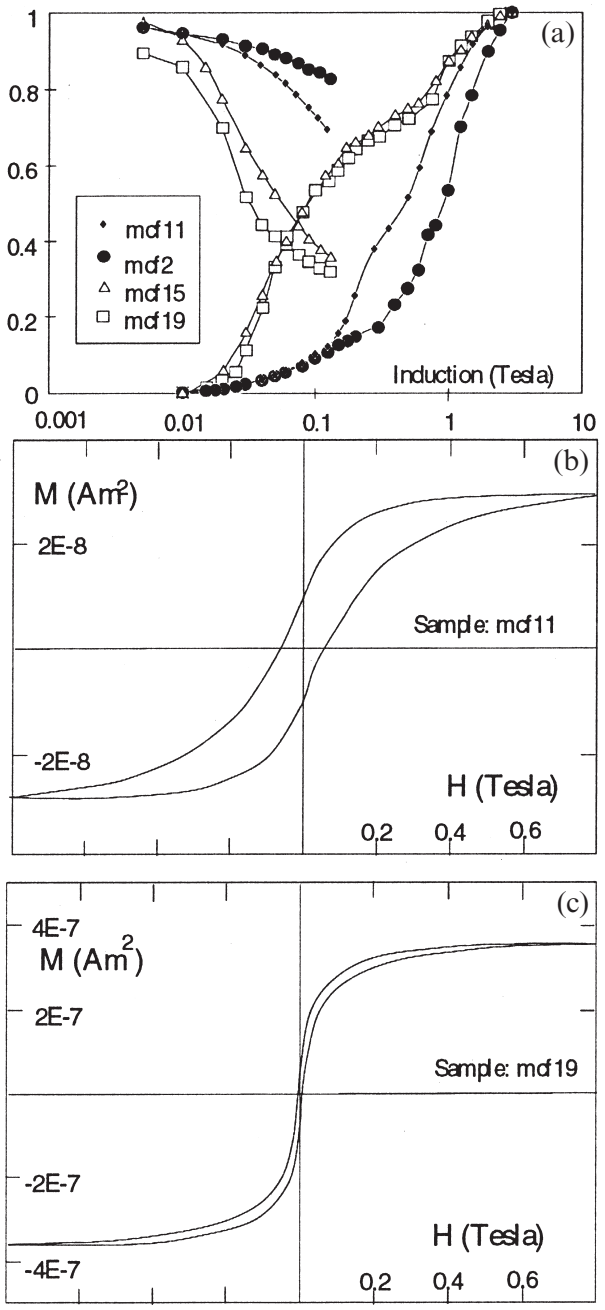


Figure 3. (a) Acquisition and alternating field (AF) demagnetization of IRM (isothermal remanent magnetization) in selected specimens. Note that AF demagnetization removes a significant fraction of the IRM in samples from sites mcf15 and mcf19; (b) and (c) are hysteresis curves of representative samples from sites mcf11 and mcf19, respectively. The large contribution of magnetite in sample mcf19 is obvious in the low coercivity and higher intensity magnetization.

of the reverse-polarity ChRM and the recent overprint may partially overlap, with intermediate directions resulting from composite magnetizations. Alternatively, the complex record of the reversal can be explained via the 'long-term' chemical remanent magnetization acquisition model of Larson *et al.* (1981), with intermediate directions resulting from the acquisition of magnetizations of normal and reverse polarity by grains of similar size and therefore similar unblocking temperature. The record suggests that the interval of reverse polarity is of short duration.

ROCK MAGNETISM

Isothermal remanent magnetization (IRM), AF demagnetization of the IRM, and hysteresis curves suggest that magnetic phases of both low and high coercive force are present in rocks of the Dewey Lake Formation, although the high-coercivity phase dominates the signal (Fig. 3a). All samples fail to reach saturation at inductions of 3 T. Samples for sites with weak overprints and well-defined ChRM are fine-grained sandstones of greater compositional and textural maturity. Specular haematite grains are concentrated along heavy-mineral laminations and show evidence of high-temperature exsolution textures, indicating that they originated from crystalline rocks and are detrital in origin (Fig. 5). Well-rounded martite grains are common. Detrital ilmenite grains have been extensively replaced by haematite. Some of the large specularite grains have experienced intense dissolution. Demagnetization behaviour and rock magnetism for samples from sites with weak overprints (high maximum unblocking temperature and high coercivity) suggest that the carrier of both the characteristic magnetization and the overprint is haematite. Petrographic inspection reveals that haematite pigment in most samples is generally observed to postdate the formation of quartz overgrowths and to predate carbonate cement. Haematite pigment is abundant as a coating of quartz grains.

For sites where the recent overprint is more developed (for example sites mcf15 and mcf19), contributions from magnetite to the IRM are more important than at sites where this component is small (e.g. sites mcf2 and mcf11). In samples from mcf15 and mcf19, for example, up to 60 per cent of the saturation IRM is removed with AF demagnetization. Furthermore, hysteresis curves for fragments from samples with a well-developed overprint may be dominated by contributions from a low-coercivity phase (Fig. 3c). Sites where a recent overprint is well developed are compositionally immature sandstones, with abundant lithic and feldspar grains (Fig. 5a). Observation of polished thin sections suggests that the magnetite crystals are present within lithic grains. These crystals of magnetite probably carry a viscous remanence. Detrital haematite grains in these samples are not abundant and show less abrasion and/or leaching than grains in samples with a well-developed high laboratory unblocking temperature ChRM.

We infer that in samples where magnetite is relatively abundant, incomplete oxidation during or soon after deposition resulted in a weak primary chemical remanent magnetization (CRM), and recent weathering resulted in acquisition of a young (north-directed and steep) CRM. The development of a CRM overprint in samples with a stable ChRM was possibly limited by the availability of Fe^{2+} ions prior to recent exposure to weathering.

PALAEOMAGNETIC POLE

Most sites where a ChRM is recorded yield moderate within-site dispersions (Table 1). High values for the α_{95} cone of confidence can be attributed in most cases to the relatively low number of samples included in each site mean calculation.

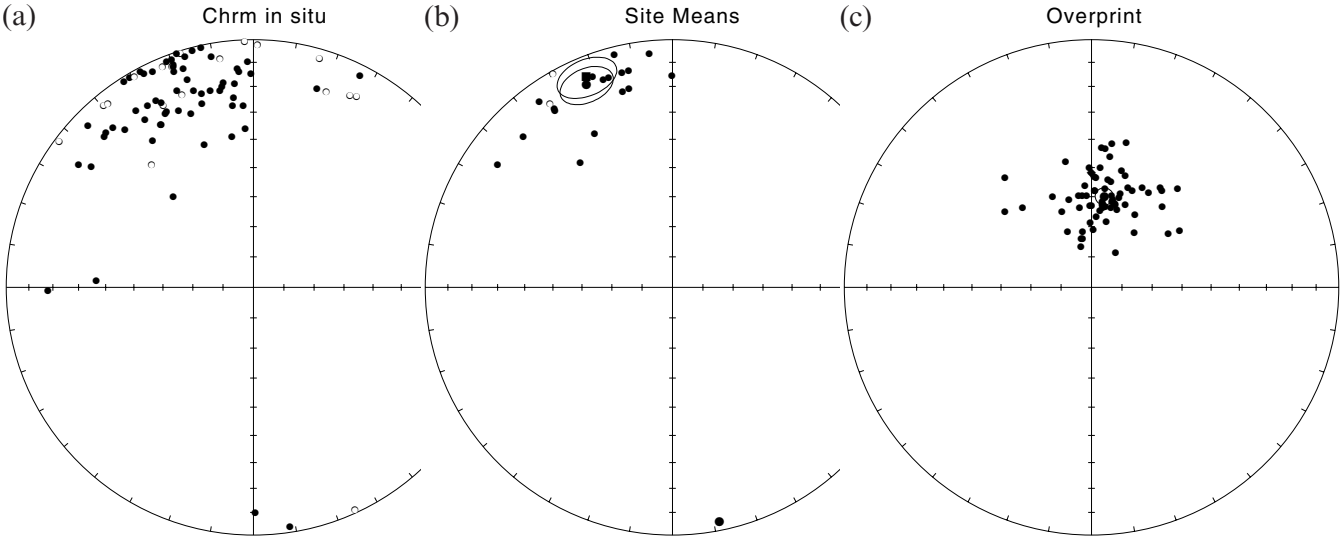


Figure 4. Equal-area stereographic projection of the characteristic magnetization isolated in samples from the Dewey Lake Formation (a). Also shown are projections of the site mean directions (b) and the low-temperature overprint removed by demagnetization techniques (c).

Table 1. Palaeomagnetic data and statistical parameters.

Site	n/nd	dec(°)	inc(°)	k	$\alpha 95$
mcf01	8/10	346.0	3.2	48.3	8.1
mcf02	6/6	330.8	−1.2	39.5	10.8
mcf04*	6/7	354.3	5.6	7.6	26.0
mcf05*	2/7	359.9	15.2		
mcf06	5/7	345.7	19.5	72.2	9.1
mcf07	4/6	341.7	12.3	39.1	14.9
mcf08*	4/6	348.6	11.3	4.4	49.6
mcf10*	1/5	326.5	15.2		
mcf11	6/6	346.8	11.6	49.2	9.6
mcf13–14	3/6	168.7	3.9	35.5	21.0
mcf14	3/6	324.4	8.2	21.6	27.2
mcf15	5/8	326.3	−11.8	14.3	21.0
mcf17*	1/5	305.2	14.5		
mcf18	3/6	328.2	15.3	10.2	18.8
mcf19	4/7	347.6	18.9	19.1	21.6
mcf20	3/4	326.5	26.8	26.4	24.5
mcf21*	1/2	315.3	15.3		
mcf22	5/7	339.3	9.5	18.8	18.1
mcf23*	5/5	343.1	12.2	3.3	49.7
mcf24*	5/4	323.6	37.2	6.8	37.9
Overall	**20**	**336.9**	**12.2**	**21.8**	**7.2**
Selected	**12**	**337.7**	**9.2**	**31.6**	**7.8**

Here n/d is the number of samples demagnetized/used in calculations k and $\alpha 95$ are the precision parameter and cone of confidence of Fisher's statistics. An asterisk indicates sites excluded from the final calculation.

Four sites were excluded from the final pole calculation because their site mean directions are poorly defined ($k < 10$ and $\alpha_{95} > 20°$). Only sites with three or more samples were included in the final pole calculation, which resulted in the exclusion of an additional four sites. The mean of the 12 remaining sites ($D = 337.7°$, $I = +9.2°$; $\alpha_{95} = 7.8°$) yields a pole at 55.2°N, 117.5°E, in excellent agreement with published Late Permian poles for the North America craton (Van der Voo 1990). The pole is also similar to the pole obtained from Dewey Lake strata in northwest Texas (Molina-Garza *et al.* 1989), located at 51.4°N, 126.2°E. The mean North America reference pole compares well with the European Late Permian pole rotated to North American coordinates using the rotation parameters of a 'tight' Bullard-type fit (Van der Voo 1990).

MAGNETOSTRATIGRAPHY

The Dewey Lake section defines a polarity sequence formed by two relatively thick normal-polarity zones (DLa+ and DLc+) separated by a thin reverse-polarity (DLb–) zone (Fig. 6). This magnetostratigraphy is consistent with data obtained for these rocks in west Texas (Molina-Garza *et al.* 1989). Although sites from the Texas localities are more broadly spaced in the section, at Caprock Canyon State Park all sites yielded normal-polarity magnetization, and at Palo Duro Canyon reverse polarities were observed between more dominant intervals of normal polarity. Reverse-polarity magnetizations observed at the base of the Dewey Lake section in west Texas were not found in the Maroon Cliffs section. Thus the Dewey Lake strata sampled in the present study are presumably older than 251 Ma.

None of the ash layers described within the Dewey Lake sequence in west Texas was recognized in the Maroon Cliffs section. Nevertheless, natural gamma logs obtained from an exploratory hole drilled at the WIPP radioactive waste repository, about 15 km to the south (Fig. 1), would suggest that the lowermost ash layer is located about 20 m above the stratigraphically highest site drilled at Maroon Cliffs (Fig. 6).

Of the entire Ochoan section preserved in southwestern North America, the Rustler Formation offers the best biostratigraphic information. An extensive marine invertebrate fauna (bivalves, brachiopods, and gastropods) from the Virginia Draw Member is Late Permian (Walter 1953). This member contains an extensive conodont fauna (Croft 1978) that Wardlaw & Grant (1992) considered to be of early Changxingian (latest Permian) age. A similar conodont fauna has also been recovered from the Culebra and lower Tamarisk members of the Rustler Formation (Croft 1978).

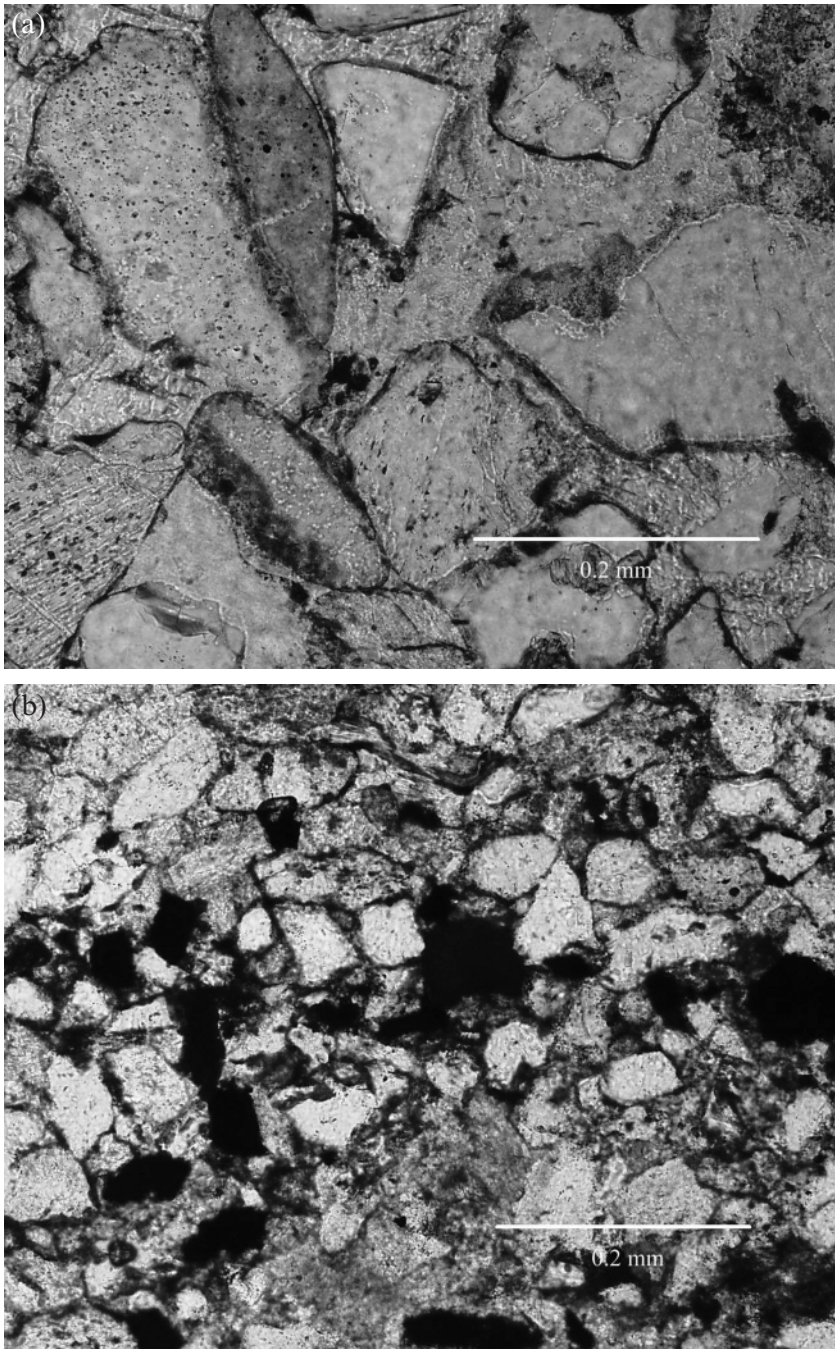


Figure 5. Digital photomicrographs of polished thin sections (plane light) from (a) a coarse-grained sample with a well-developed overprint, and (b) a sample with a weak overprint and a well-developed characteristic magnetization, showing detrital specularite grains laminations. Amplification in both images is 20×.

Correlation of the short polarity zonation of the Maroon Cliffs section of the Dewey Lake Formation with the polarity stratigraphy of marine sections in China and west Asia is reasonably unambiguous. Based on the biostratigraphic and geochronological data (Renne *et al.* 1996), the Dewey Lake magnetostratigraphy is consistent with deposition during the late Changxingian. Magnetostratigraphic correlation across the Permian–Triassic boundary (Fig. 7) is still problematic because of the rather low reliability of the magnetostratigraphic record from sections in the South China block, where polarity was typically interpreted from demagnetization trajectories for samples collected from sparsely sampled sections (Steiner *et al.* 1989; Heller *et al.* 1988). Nonetheless, apparently long intervals of normal polarity in Changxingian or age-equivalent strata have been observed in the Russian platform, Pakistan, South China, and North China. A short reverse-polarity zone within the predominantly normal-polarity interval has also been observed in several of the sections (Fig. 7). This further supports the correlation of Dewey Lake rocks with the Changxingian segment of the Permian–Triassic polarity sequence.

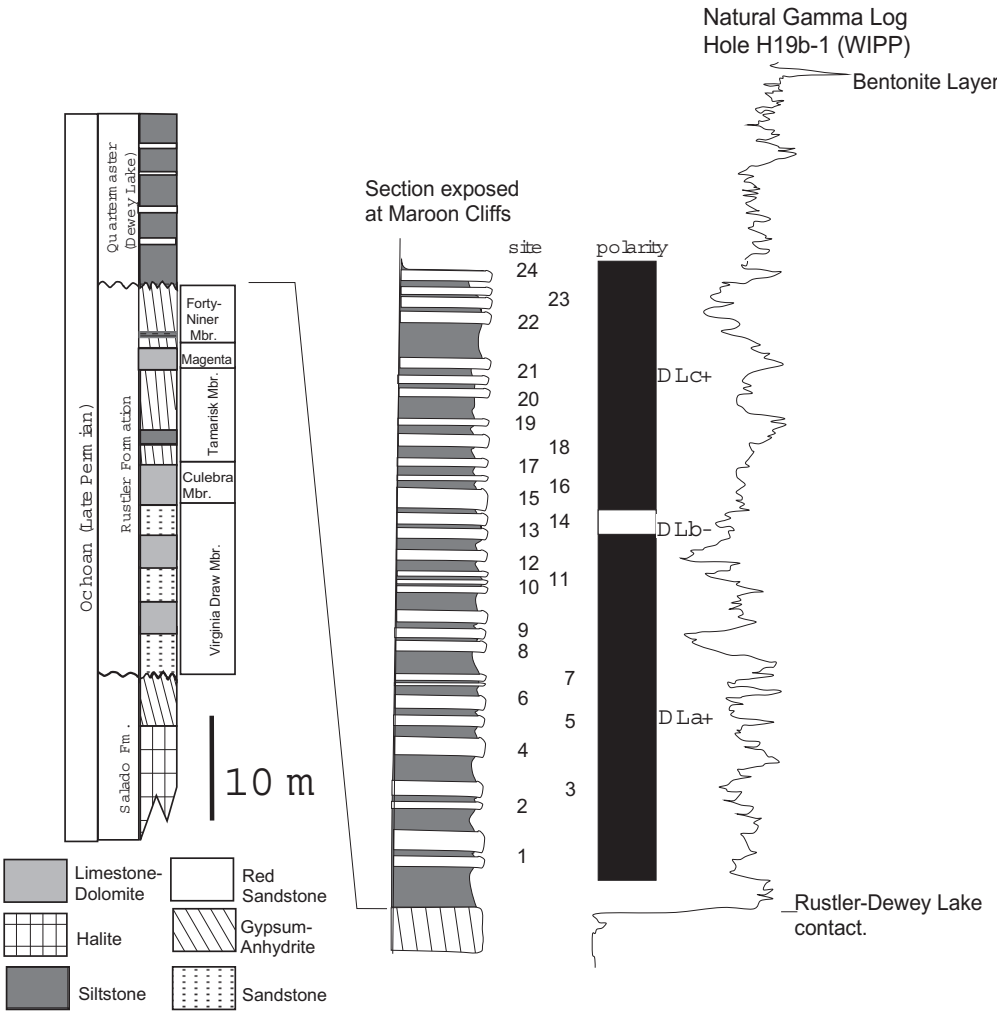


Figure 6. Schematic geological column of the Ochoan succession of southeast New Mexico (left). Also shown is a stratigraphic section of the Dewey Lake Formation at Maroon Cliffs with the location of sampling sites and interpretation of magnetic polarity (centre). On the far right we show a natural gamma log for a hole drilled by the US Geological Survey about 15 km south of the sampled section, where we interpret the position of a bentonite layer about 20 m above site mcf24.

The Late Permian is an interval characterized by mixed polarities, often referred to as the Illawarra Zone, which immediately followed the end of the reverse-polarity Kiaman Superchron (Irving & Parry 1963; Opdyke 1995). The term Illawarra is inadequate as a magnetostratigraphic unit (magnetozone) because its termination cannot be defined. On the other hand, the end of the Kiaman Superchron is a global stratigraphic marker that allows intercontinental correlation of marine and non-marine biochronologies and stratigraphic successions. Workers have determined the stratigraphic position of the end of the Kiaman Superchron in lower Tatarian strata of the Russian platform (Menning 1995; Kotlyar *et al.* 1986), and Murgabian strata in the Salt Range (Haag & Heller 1991). It also has been suggested, but not confirmed, that upper Guadalupian strata in southwest North America record the end of the Kiaman Superchron (Peterson & Nairn 1971). Work in North China (Embleton *et al.* 1996) has suggested an older age for the termination of the Kiaman Superchron, in the Upper Permian Ufimian stage. Menning & Jin (1998) explain this apparent discrepancy by erroneous correlation of the Shihezi Formation, a problem that arises from the inadequate correlation between the timescale of Harland *et al.* (1990) and the Chinese Permian regional chronostratigraphic scale. Using a revised stratigraphy, Menning & Jin (1998) suggest that the end of the Kiaman Superchron in Chinese strata is in the late Maokouan stage, in agreement with the Russian platform data. Although Guadalupian sections in North America have not been sampled in sufficient detail, carbonate rocks of the San Andres Formation in central New Mexico contain normal-polarity magnetizations of Permian age (Molina-Garza & Geissman 1999).

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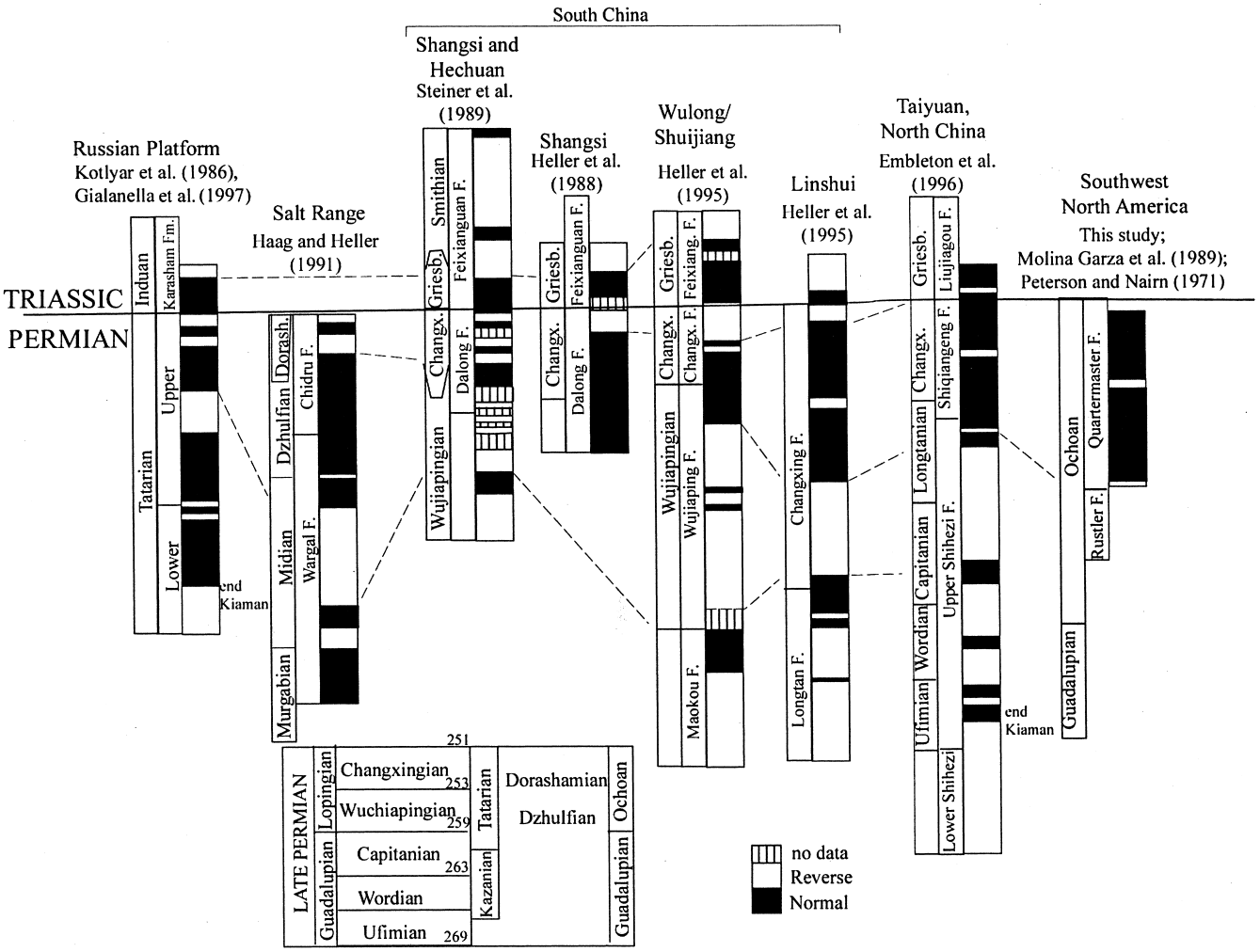


Figure 7. Summary of inferred magnetostratigraphy across the Permian–Triassic boundary, modified after Heller *et al.* (1995).

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